

```
mysql> /* Date and time functions
/*> 1.curdate()-->current date.
/*> 2.curtime()-->current time.
/*> 3.now()-->current date and time.
/*> 4.date()-->extracted date part.
/*> 5.time()-->extracted time part.
/*> 6.year()-->extract year.
/*> 7.month()-->extract month.
/*> 8.day()/dayofmonth()--> extract day of the month.
/*> 9.week()--> extract week number.
/*> 10.dayofweek()-->extract day of the week.
/*> 11.date_add()--> add time to a date.
/*> 12.date_sub()--> subtract time from a date.
/*> 13.datediff()-->difference between two dates.
/*> 14.timestampdiff()-->difference between two 'Datetime' values. */
```

```
mysql>
mysql> select curdate() as current__date;
```

```
+-----+
| current__date |
+-----+
| 2026-09-16    |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select curtime() as current__time;
```

```
+-----+
| current__time |
+-----+
| 14:21:01      |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select now() as current_date_time;
```

```
+-----+
| current_date_time |
+-----+
| 2026-09-16 14:21:33 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select sysdate();
```

```
+-----+
| sysdate()        |
+-----+
| 2026-09-16 14:21:47 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select date(now()) as extracted_date;
```

```
+-----+
```

```
| extracted_date |
+-----+
| 2026-09-16    |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select time(now()) as extracted_time;
+-----+
| extracted_time |
+-----+
| 14:23:52       |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select year(now()) as current__year;
+-----+
| current__year |
+-----+
|          2026 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select month(now())as month_part;
+-----+
| month_part |
+-----+
|          9 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select day(now()) as day_part;
+-----+
| day_part |
+-----+
|        16 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select week(now()) as current__week;
+-----+
| current__week |
+-----+
|          37 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select dayofweek(now())as current__day;
+-----+
| current__day |
+-----+
```

```
|          4 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select date_add(now(),interval 100 day) as new_date;
```

```
+-----+
| new_date          |
+-----+
| 2026-12-25 14:30:09 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select date_sub(now(),interval 100 day) as new_date;
```

```
+-----+
| new_date          |
+-----+
| 2026-06-08 14:30:26 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select date_add(now(),interval 365 day) as new_date;
```

```
+-----+
| new_date          |
+-----+
| 2027-09-16 14:31:02 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select datediff(now(),'2024-03-02')as days_diff;
```

```
+-----+
| days_diff |
+-----+
|          928 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select datediff(now(),'2024-09-16')as days_diff;
```

```
+-----+
| days_diff |
+-----+
|          730 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select str_to_date('2024-03-20','%Y-%m-%d')as formatted_date;
```

```
+-----+
| formatted_date |
+-----+
| 2024-03-20    |
+-----+
```

1 row in set (0.00 sec)

```
mysql> select date_format(now(), '%W,%m,%d,%Y') as formatted_date;
```

formatted_date
Wednesday,09,16,2026

1 row in set (0.00 sec)

```
mysql> select last_day(now()) as last_day_of_month;
```

last_day_of_month
2026-09-30

1 row in set (0.00 sec)

```
mysql> select makedate(2024,60) as result_date;
```

result_date
2024-02-29

1 row in set (0.03 sec)

```
mysql> select makedate(2025,60) as result_date;
```

result_date
2025-03-01

1 row in set (0.00 sec)

```
mysql> select maketime(14,25,12) as result_time;
```

result_time
14:25:12

1 row in set (0.00 sec)

```
mysql> select monthname(now()) as current__month;
```

current__month
September

1 row in set (0.00 sec)

```
mysql> select dayname(now())as day_name;
```

```
+-----+  
| day_name |  
+-----+  
| Wednesday |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select quarter(now())as quarter_number;
```

```
+-----+  
| quarter_number |  
+-----+  
| 3 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select sec_to_time(3661)as time_value;
```

```
+-----+  
| time_value |  
+-----+  
| 01:01:01 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select time_to_sec('01:01:01')as sec_value;
```

```
+-----+  
| sec_value |  
+-----+  
| 3661 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> CREATE TABLE employee_attendance
```

```
-> (  
-> emp_id INT PRIMARY KEY AUTO_INCREMENT,  
-> emp_name VARCHAR(100),  
-> department VARCHAR(50),  
-> joining_date DATE,  
-> birth_date DATE,  
-> login_time DATETIME,  
-> logout_time DATETIME,  
-> salary DECIMAL(10,2)  
-> );
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql>
```

```
mysql> INSERT INTO employee_attendance
```

```
-> (emp_name, department, joining_date, birth_date, login_time, logout_time,  
salary)  
-> VALUES
```

```

-> ('Rahul','IT','2021-06-15','1998-05-10','2024-11-20 09:05:10','2024-11-20
18:15:30',55000),
-> ('Sneha','HR','2020-03-21','1996-11-18','2024-11-21 08:55:20','2024-11-21
17:45:50',48000),
-> ('Arjun','Finance','2019-08-10','1994-07-30','2024-11-22
09:15:45','2024-11-22 18:35:20',62000),
-> ('Priya','IT','2022-01-05','2000-01-25','2024-11-23 09:00:00','2024-11-23
17:55:40',51000),
-> ('Kiran','Admin','2018-12-11','1992-09-15','2024-11-24 08:40:30','2024-11-24
17:30:00',45000),
-> ('Anjali','Testing','2023-04-17','1999-12-20','2024-11-25
09:12:10','2024-11-25 18:05:00',47000),
-> ('Vijay','IT','2021-09-14','1997-04-08','2024-11-26 09:08:15','2024-11-26
18:20:40',59000),
-> ('Divya','HR','2019-10-19','1995-06-11','2024-11-27 08:50:30','2024-11-27
17:42:15',50000),
-> ('Ramesh','Finance','2017-07-01','1991-08-14','2024-11-28
09:03:00','2024-11-28 18:12:00',70000),
-> ('Pooja','Testing','2022-05-09','2001-03-19','2024-11-29
09:18:50','2024-11-29 18:25:10',53000);
Query OK, 10 rows affected (0.02 sec)
Records: 10 Duplicates: 0 Warnings: 0

```

```
mysql> select * from employee_attendance;
```

emp_id	emp_name	department	joining_date	birth_date	login_time	logout_time	salary
1	Rahul	IT	2021-06-15	1998-05-10	2024-11-20 09:05:10	2024-11-20 18:15:30	55000.00
2	Sneha	HR	2020-03-21	1996-11-18	2024-11-21 08:55:20	2024-11-21 17:45:50	48000.00
3	Arjun	Finance	2019-08-10	1994-07-30	2024-11-22 09:15:45	2024-11-22 18:35:20	62000.00
4	Priya	IT	2022-01-05	2000-01-25	2024-11-23 09:00:00	2024-11-23 17:55:40	51000.00
5	Kiran	Admin	2018-12-11	1992-09-15	2024-11-24 08:40:30	2024-11-24 17:30:00	45000.00
6	Anjali	Testing	2023-04-17	1999-12-20	2024-11-25 09:12:10	2024-11-25 18:05:00	47000.00
7	Vijay	IT	2021-09-14	1997-04-08	2024-11-26 09:08:15	2024-11-26 18:20:40	59000.00
8	Divya	HR	2019-10-19	1995-06-11	2024-11-27 08:50:30	2024-11-27 17:42:15	50000.00
9	Ramesh	Finance	2017-07-01	1991-08-14	2024-11-28 09:03:00	2024-11-28 18:12:00	70000.00
10	Pooja	Testing	2022-05-09	2001-03-19	2024-11-29 09:18:50	2024-11-29 18:25:10	53000.00

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> select emp_name,date(login_time)as login_time_only from employee_attendance;
```

```
+-----+-----+
| emp_name | login_time_only |
+-----+-----+
| Rahul    | 2024-11-20      |
| Sneha    | 2024-11-21      |
| Arjun    | 2024-11-22      |
| Priya    | 2024-11-23      |
| Kiran    | 2024-11-24      |
| Anjali   | 2024-11-25      |
| Vijay    | 2024-11-26      |
| Divya    | 2024-11-27      |
| Ramesh   | 2024-11-28      |
| Pooja    | 2024-11-29      |
+-----+-----+
```

```
10 rows in set (0.00 sec)
```

```
mysql> select emp_name,time(login_time)as login_time_only from employee_attendance;
```

```
+-----+-----+
| emp_name | login_time_only |
+-----+-----+
| Rahul    | 09:05:10        |
| Sneha    | 08:55:20        |
| Arjun    | 09:15:45        |
| Priya    | 09:00:00        |
| Kiran    | 08:40:30        |
| Anjali   | 09:12:10        |
| Vijay    | 09:08:15        |
| Divya    | 08:50:30        |
| Ramesh   | 09:03:00        |
| Pooja    | 09:18:50        |
+-----+-----+
```

```
10 rows in set (0.00 sec)
```

```
mysql> select emp_name,year(joining_date)as joining_year from employee_attendance;
```

```
+-----+-----+
| emp_name | joining_year |
+-----+-----+
| Rahul    | 2021         |
| Sneha    | 2020         |
| Arjun    | 2019         |
| Priya    | 2022         |
| Kiran    | 2018         |
| Anjali   | 2023         |
| Vijay    | 2021         |
| Divya    | 2019         |
+-----+-----+
```

Ramesh	2017
Pooja	2022

10 rows in set (0.00 sec)

```
mysql> select emp_name,month(birth_date) as birth_month_number from employee_attendance;
```

emp_name	birth_month_number
Rahul	5
Sneha	11
Arjun	7
Priya	1
Kiran	9
Anjali	12
Vijay	4
Divya	6
Ramesh	8
Pooja	3

10 rows in set (0.00 sec)

```
mysql> select emp_name,day(birth_date) as day_of_birth from employee_attendance;
```

emp_name	day_of_birth
Rahul	10
Sneha	18
Arjun	30
Priya	25
Kiran	15
Anjali	20
Vijay	8
Divya	11
Ramesh	14
Pooja	19

10 rows in set (0.00 sec)

```
mysql> select emp_name,week(joining_date) as week_number from employee_attendance;
```

emp_name	week_number
Rahul	24
Sneha	11
Arjun	31
Priya	1
Kiran	49
Anjali	16

Vijay	37
Divya	41
Ramesh	26
Pooja	19

10 rows in set (0.00 sec)

```
mysql> select emp_name,dayofweek(joining_date) as week_day from
employee_attendance;
```

emp_name	week_day
Rahul	3
Sneha	7
Arjun	7
Priya	4
Kiran	3
Anjali	2
Vijay	3
Divya	7
Ramesh	7
Pooja	2

10 rows in set (0.00 sec)

```
mysql> select emp_name,date_add(joining_date,interval 30 day) as new_date from
employee_attendance;
```

emp_name	new_date
Rahul	2021-07-15
Sneha	2020-04-20
Arjun	2019-09-09
Priya	2022-02-04
Kiran	2019-01-10
Anjali	2023-05-17
Vijay	2021-10-14
Divya	2019-11-18
Ramesh	2017-07-31
Pooja	2022-06-08

10 rows in set (0.00 sec)

```
mysql> select emp_name,date_sub(joining_date,interval 15 day) as new_date from
employee_attendance;
```

emp_name	new_date
Rahul	2021-05-31
Sneha	2020-03-06

Arjun	2019-07-26
Priya	2021-12-21
Kiran	2018-11-26
Anjali	2023-04-02
Vijay	2021-08-30
Divya	2019-10-04
Ramesh	2017-06-16
Pooja	2022-04-24

10 rows in set (0.00 sec)

```
mysql> select emp_name,datediff(joining_date,now()) as worked_days from
employee_attendance;
```

emp_name	worked_days
Rahul	-1919
Sneha	-2370
Arjun	-2594
Priya	-1715
Kiran	-2836
Anjali	-1248
Vijay	-1828
Divya	-2524
Ramesh	-3364
Pooja	-1591

10 rows in set (0.00 sec)

```
mysql> select emp_name,datediff(joining_date,logout_time) as worked_days from
employee_attendance;
```

emp_name	worked_days
Rahul	-1254
Sneha	-1706
Arjun	-1931
Priya	-1053
Kiran	-2175
Anjali	-588
Vijay	-1169
Divya	-1866
Ramesh	-2707
Pooja	-935

10 rows in set (0.00 sec)

```
mysql> select emp_name,datediff(login_time,logout_time) as worked_days from
employee_attendance;
```

emp_name	worked_days
----------	-------------

Rahul	0
Sneha	0
Arjun	0
Priya	0
Kiran	0
Anjali	0
Vijay	0
Divya	0
Ramesh	0
Pooja	0

10 rows in set (0.00 sec)

```
mysql> select emp_name,date_format(joining_date, '%W,%m,%d,%Y')as work_hours from employee_attendance;
```

emp_name	work_hours
Rahul	Tuesday,06,15,2021
Sneha	Saturday,03,21,2020
Arjun	Saturday,08,10,2019
Priya	Wednesday,01,05,2022
Kiran	Tuesday,12,11,2018
Anjali	Monday,04,17,2023
Vijay	Tuesday,09,14,2021
Divya	Saturday,10,19,2019
Ramesh	Saturday,07,01,2017
Pooja	Monday,05,09,2022

10 rows in set (0.00 sec)

```
mysql> select emp_name,year(now()) as current_joined from employee_attendance;
```

emp_name	current_joined
Rahul	2026
Sneha	2026
Arjun	2026
Priya	2026
Kiran	2026
Anjali	2026
Vijay	2026
Divya	2026
Ramesh	2026
Pooja	2026

10 rows in set (0.00 sec)

```
mysql> select emp_name,joining_date from employee_attendance where year(joining_date)=year(curdate());
```

Empty set (0.00 sec)

```
mysql> select * from employee_attendance;
```

emp_id	emp_name	department	joining_date	birth_date	login_time
logout_time	salary				
1	Rahul	IT	2021-06-15	1998-05-10	2024-11-20 09:05:10
2024-11-20 18:15:30	55000.00				
2	Sneha	HR	2020-03-21	1996-11-18	2024-11-21 08:55:20
2024-11-21 17:45:50	48000.00				
3	Arjun	Finance	2019-08-10	1994-07-30	2024-11-22 09:15:45
2024-11-22 18:35:20	62000.00				
4	Priya	IT	2022-01-05	2000-01-25	2024-11-23 09:00:00
2024-11-23 17:55:40	51000.00				
5	Kiran	Admin	2018-12-11	1992-09-15	2024-11-24 08:40:30
2024-11-24 17:30:00	45000.00				
6	Anjali	Testing	2023-04-17	1999-12-20	2024-11-25 09:12:10
2024-11-25 18:05:00	47000.00				
7	Vijay	IT	2021-09-14	1997-04-08	2024-11-26 09:08:15
2024-11-26 18:20:40	59000.00				
8	Divya	HR	2019-10-19	1995-06-11	2024-11-27 08:50:30
2024-11-27 17:42:15	50000.00				
9	Ramesh	Finance	2017-07-01	1991-08-14	2024-11-28 09:03:00
2024-11-28 18:12:00	70000.00				
10	Pooja	Testing	2022-05-09	2001-03-19	2024-11-29 09:18:50
2024-11-29 18:25:10	53000.00				

10 rows in set (0.00 sec)

```
mysql> select emp_name,birth_date from employee_attendance where  
month(birth_date)=10;
```

Empty set (0.00 sec)

```
mysql> select emp_name,birth_date from employee_attendance where  
day(birth_date)=10;
```

emp_name	birth_date
Rahul	1998-05-10

1 row in set (0.00 sec)

```
mysql> select emp_name,joining_date from employee_attendance where  
month(joining_date)='june';
```

Empty set, 1 warning (0.00 sec)

```
mysql> select emp_name,joining_date from employee_attendance where  
monthname(joining_date)='june';
```

```
+-----+-----+  
| emp_name | joining_date |  
+-----+-----+  
| Rahul    | 2021-06-15   |  
+-----+-----+  
1 row in set (0.00 sec)
```

```
mysql> select emp_name,birth_date from employee_attendance where  
day(birth_date)=10;
```

```
+-----+-----+  
| emp_name | birth_date |  
+-----+-----+  
| Rahul    | 1998-05-10 |  
+-----+-----+  
1 row in set (0.00 sec)
```

```
mysql> select emp_name,joining_date from employee_attendance where  
quarter(joining_date)=2;
```

```
+-----+-----+  
| emp_name | joining_date |  
+-----+-----+  
| Rahul    | 2021-06-15   |  
| Anjali   | 2023-04-17   |  
| Pooja    | 2022-05-09   |  
+-----+-----+  
3 rows in set (0.00 sec)
```

```
mysql> select emp_name,timestampdiff(year,birth_date,curdate())as age from  
employee_attendance;
```

```
+-----+-----+  
| emp_name | age |  
+-----+-----+  
| Rahul    | 28 |  
| Sneha    | 29 |  
| Arjun    | 32 |  
| Priya    | 26 |  
| Kiran    | 34 |  
| Anjali    | 26 |  
| Vijay    | 29 |  
| Divya    | 31 |  
| Ramesh   | 35 |  
| Pooja    | 25 |  
+-----+-----+  
10 rows in set (0.00 sec)
```

```
mysql> select emp_name from employee_attendance where  
timestampdiff(year,joining_date,curdate())>3;
```

emp_name
Rahul
Sneha
Arjun
Priya
Kiran
Vijay
Divya
Ramesh
Pooja

9 rows in set (0.00 sec)

```
mysql> select emp_name,timestampdiff(hour,login_time,logout_time)as working_hours
from employee_attendance;
```

emp_name	working_hours
Rahul	9
Sneha	8
Arjun	9
Priya	8
Kiran	8
Anjali	8
Vijay	9
Divya	8
Ramesh	9
Pooja	9

10 rows in set (0.00 sec)

```
mysql> select emp_name,timestampdiff(minute,login_time,logout_time)as
working_minutes from employee_attendance;
```

emp_name	working_minutes
Rahul	550
Sneha	530
Arjun	559
Priya	535
Kiran	529
Anjali	532
Vijay	552
Divya	531
Ramesh	549
Pooja	546

10 rows in set (0.00 sec)

```
mysql> select emp_name,login_time from employee_attendance where
time(login_time)<'9:00:00';
```

emp_name	login_time
Rahul	2024-11-20 09:05:10
Sneha	2024-11-21 08:55:20
Arjun	2024-11-22 09:15:45
Priya	2024-11-23 09:00:00
Kiran	2024-11-24 08:40:30
Anjali	2024-11-25 09:12:10
Vijay	2024-11-26 09:08:15
Divya	2024-11-27 08:50:30
Ramesh	2024-11-28 09:03:00
Pooja	2024-11-29 09:18:50

10 rows in set (0.00 sec)

```
mysql> select emp_name,login_time from employee_attendance where
time(login_time)>'9:10:00';
Empty set (0.00 sec)
```

```
mysql> select emp_name,login_time from employee_attendance where
time(login_time)<'9:00:00';\
```

emp_name	login_time
Rahul	2024-11-20 09:05:10
Sneha	2024-11-21 08:55:20
Arjun	2024-11-22 09:15:45
Priya	2024-11-23 09:00:00
Kiran	2024-11-24 08:40:30
Anjali	2024-11-25 09:12:10
Vijay	2024-11-26 09:08:15
Divya	2024-11-27 08:50:30
Ramesh	2024-11-28 09:03:00
Pooja	2024-11-29 09:18:50

10 rows in set (0.00 sec)

```
mysql> select emp_name, login_time from employee_attendance where
time(login_time)<'9:00:00';
```

emp_name	login_time
Rahul	2024-11-20 09:05:10
Sneha	2024-11-21 08:55:20
Arjun	2024-11-22 09:15:45
Priya	2024-11-23 09:00:00

Kiran	2024-11-24 08:40:30
Anjali	2024-11-25 09:12:10
Vijay	2024-11-26 09:08:15
Divya	2024-11-27 08:50:30
Ramesh	2024-11-28 09:03:00
Pooja	2024-11-29 09:18:50

-----+
10 rows in set (0.00 sec)

```
mysql> select emp_name,login_time from employee_attendance where
time(login_time)<'9:00:00';
```

emp_name	login_time
Rahul	2024-11-20 09:05:10
Sneha	2024-11-21 08:55:20
Arjun	2024-11-22 09:15:45
Priya	2024-11-23 09:00:00
Kiran	2024-11-24 08:40:30
Anjali	2024-11-25 09:12:10
Vijay	2024-11-26 09:08:15
Divya	2024-11-27 08:50:30
Ramesh	2024-11-28 09:03:00
Pooja	2024-11-29 09:18:50

-----+
10 rows in set (0.00 sec)

```
mysql> select emp_name,login_time from employee_attendance where time(login_time)
<'9:00:00';
```

emp_name	login_time
Rahul	2024-11-20 09:05:10
Sneha	2024-11-21 08:55:20
Arjun	2024-11-22 09:15:45
Priya	2024-11-23 09:00:00
Kiran	2024-11-24 08:40:30
Anjali	2024-11-25 09:12:10
Vijay	2024-11-26 09:08:15
Divya	2024-11-27 08:50:30
Ramesh	2024-11-28 09:03:00
Pooja	2024-11-29 09:18:50

-----+
10 rows in set (0.00 sec)

```
mysql> select emp_name,joining_date from employee_attendance where
dayname(joining_date)='monday';
```

emp_name	joining_date
----------	--------------

Anjali	2023-04-17
Pooja	2022-05-09

2 rows in set (0.00 sec)

```
mysql> select emp_name,birth_date from employee_attendance where
monthname(birth_date) like 'j%';
```

emp_name	birth_date
Arjun	1994-07-30
Priya	2000-01-25
Divya	1995-06-11

3 rows in set (0.00 sec)

```
mysql> select emp_name,joining_date from employee_attendance where
week(joining_date)=25;
Empty set (0.00 sec)
```

```
mysql> select emp_name,last_day(birth_date)as last_day_birth_month from
employee_attendance;
```

emp_name	last_day_birth_month
Rahul	1998-05-31
Sneha	1996-11-30
Arjun	1994-07-31
Priya	2000-01-31
Kiran	1992-09-30
Anjali	1999-12-31
Vijay	1997-04-30
Divya	1995-06-30
Ramesh	1991-08-31
Pooja	2001-03-31

10 rows in set (0.00 sec)

```
mysql> notee
```